Syllabus & Course Overview

Machine Learning and Big Data - DATA622

CUNY School of Professional Studies

Agenda

1. Introductions
2. Syllabus
3. Grading
4. Calendar
5. Textbooks
6. Week 1 – Tasks

Syllabus

Week #	Week starting	Topics	Key Tasks	Tasks Due
1	Wed 28-Aug	Introduction to 622, Intro to Machine Learning	Discussion 1	Sun 01-Sep
2	Mon 02-Sep	Linear & Logistic Regression	Discussion 2	Sun 08-Sep
3	Mon 09-Sep	Classification: Discriminant Analysis (LDA, QDA)	Discussion 3	Sun 15-Sep
4	Mon 16-Sep	Classification: kNN, Naïve Bayes	Discussion 4	Sun 22-Sep
5	Mon 23-Sep	Tree based methods: Decision Trees	Discussion 5	Sun 29-Sep
6	Mon 30-Sep	Tree based methods: Bagging, Random Forests, Boosting	Discussion 6 + Homework 1	Sun 06-Oct
7	Mon 07-Oct	Tree based methods: Boosting and Additive Trees	Discussion 7	Sun 13-Oct
8	Mon 14-Oct	Support Vector Machines	Discussion 8	Sun 20-Oct
9	Mon 21-Oct	Support Vector Machines	Discussion 9 + Homework 2	Sun 27-Oct
10	Mon 28-Oct	Unsupervised Learning: Clustering	Discussion 10	Sun 03-Nov
11	Mon 04-Nov	Unsupervised Learning: PCA and Distance metrics	Discussion 11	Sun 10-Nov
12	Mon 11-Nov	Resampling and model selection and other topics	Discussion 12	Sun 17-Nov
13	Mon 18-Nov	Bias Variance Tradeoff	Discussion 13 + Homework 3	Sun 24-Nov
	Mon 15-Nov	Thanksgiving week (no classes)		
14	Mon 02-Dec	Neural networks continued & Foundation models	Discussion 14	Sun 08-Dec
15	Mon 09-Dec	Ethics and Responsible AI	Discussion 15 + Homework 4	Sun 15-Dec

Course Grading

Course assignments	Number of deliverables	Points for each deliverable	Total available points
Homework Assignments	4 x	100 points	= 400 points
There will be 4 homework assignments: 10% each (100 points each). These are used to used to re-enforce course concepts.			
Quizzes	8x	20 points	= 160 points
Periodic quiz			
Discussion Board	10 x	14 points	= 140 points
A student is required to participate (by introducing a question).			

700 points (100%)

Weekly calls

- Class calls will be recorded and posted online
- Thursday's 7PM ET
  Call to go over Weekly tasks and concepts.

Zoom is on invitations sent to everyone, and on Blackboard,

Textbooks

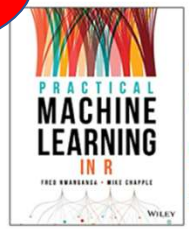
- Our text book is: Practical Machine Learning in R (2020), Fred Nwanganga, et al
- The course, syllabus and examples will focus on R
- R is the language many of you are familiar with from prerequisites
- Python is not part of this course, but I will try and accommodate you

R
(preferred)

OR

Python

$

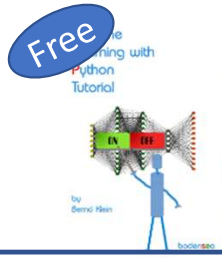


TITLE: Practical Machine Learning in R (2020)
AUTHORS: Fred Nwanganga, Mike Chapple
URL: https://www.wiley.com/en-us/Practical+Machine+Learning+in+R-p-9781119591535

Free

TITLE: Machine Learning Projects in R
SOURCE: ProjectPro
URL: https://www.projectpro.io/projects/data-science-projects/machine-learning-projects-in-r

Free



TITLE: Machine Learning with Python Tutorial (2021)
AUTHORS: Bernd Klein
URL: https://python-course.eu/books/bernd_klein_python_and_machine_learning_a4.pdf

Free

TITLE: Python Machine Learning Projects
AUTHORS: Brian Boucheron & Lisa Tagliaferri, DigitalOcean
URL: https://assets.digitalocean.com/books/python/machine-learning-projects-python.pdf

Week 1

1. Discussion Board Week 1 (introduce yourselves) – Due Sunday EOD
2. Reading: Practical Machine Learning, chapters 1 & 2
3. Concepts:
   - Supervised and unsupervised learning; machine learning vs. statistical learning
     Watch: https://youtu.be/5N9V07EIfIg (18 minutes)
     Watch: https://www.youtube.com/watch?v=Z0v9QMkA3dA (12 minutes)
   - A quick intro to reinforcement learning (20 minutes)
     Read: https://deepsense.ai/what-is-reinforcement-learning-the-complete-guide/

Week 2

1. Discussion Board Week 2
2. Reading (Practical Machine Learning with R):
   - Linear Regression: chapter 4
   - Logistic regression: chapter 5
3. Concepts:
   - Helpful lecture slides (these slides go into the topics we will be covering next week. Up to slide #28 covers this week's topic): https://web.stanford.edu/~hastie/MOOC-Slides/classification.pdf
   - Multinomial logistic regression: Please see: https://stats.idre.ucla.edu/r/dae/multinomial-logistic-regression/
4. Helpful videos:
   - Statistical learning and models (the whole playlist): https://www.youtube.com/watch?v=p9n2w236B48&list=PL5-da3qGB5IDvuFPNoSqheihPOQNJpzyy&index=1

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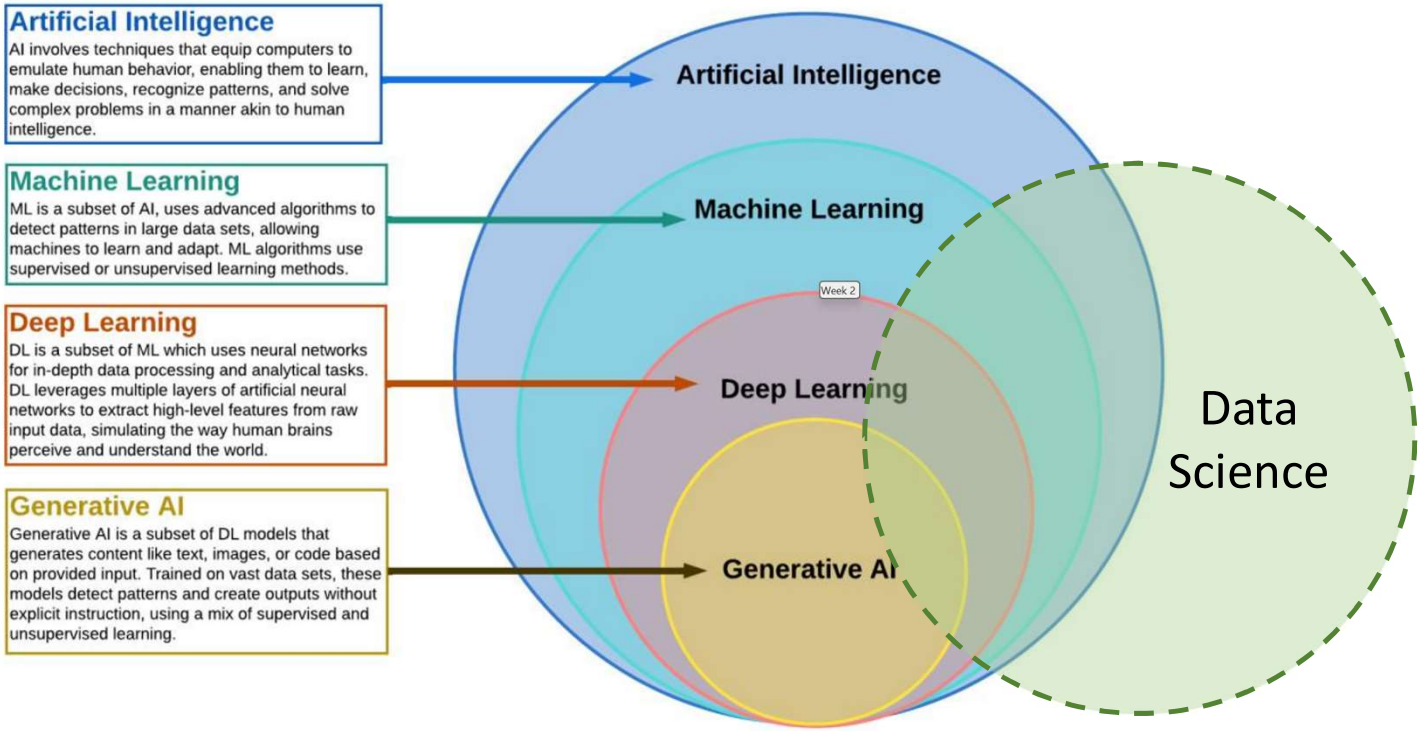
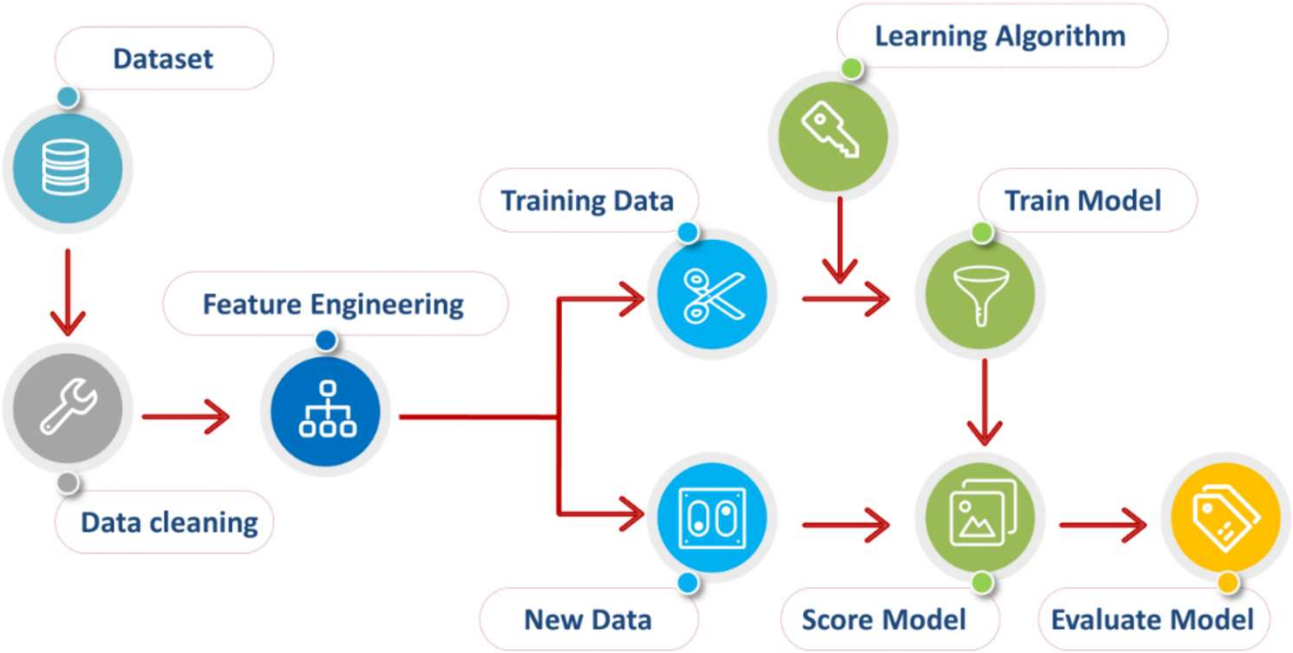
Machine Learning



Source: "The Application of AutoML Techniques in Diabetes Diagnosis: Current Approaches, Performance, and Future Directions", Zhuhadar, Lily Popova; Lytras, Miltiadis, 2023

Machine Learning Process



Source: "Neural Network for Porosity Prediction in Carbonate Formation", Huda F. Al-Saad, 2023

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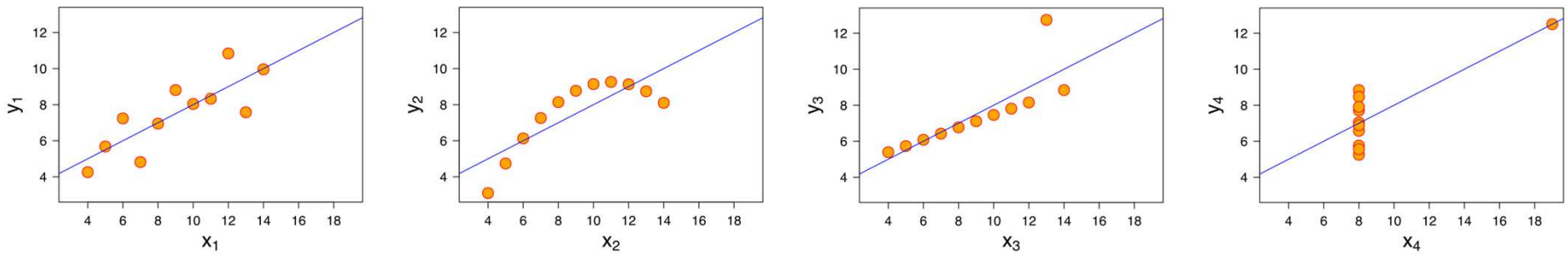
Exploratory Data Analysis (EDA)

What do these data sets have in common?



Not much. Except….

Identical means, variances, correlation, coefficients of determination, regressions

Property	Value
Mean of x	9
Sample variance of x: s_x^2	11
Mean of y	7.50
Sample variance of y: s_y^2	4.125
Correlation between x and y	0.816
Linear regression line	y = 3.00 + 0.500x
Coefficient of determination of the linear regression: R^2	0.67

Source: Anscombe's quartet from Wikipedia

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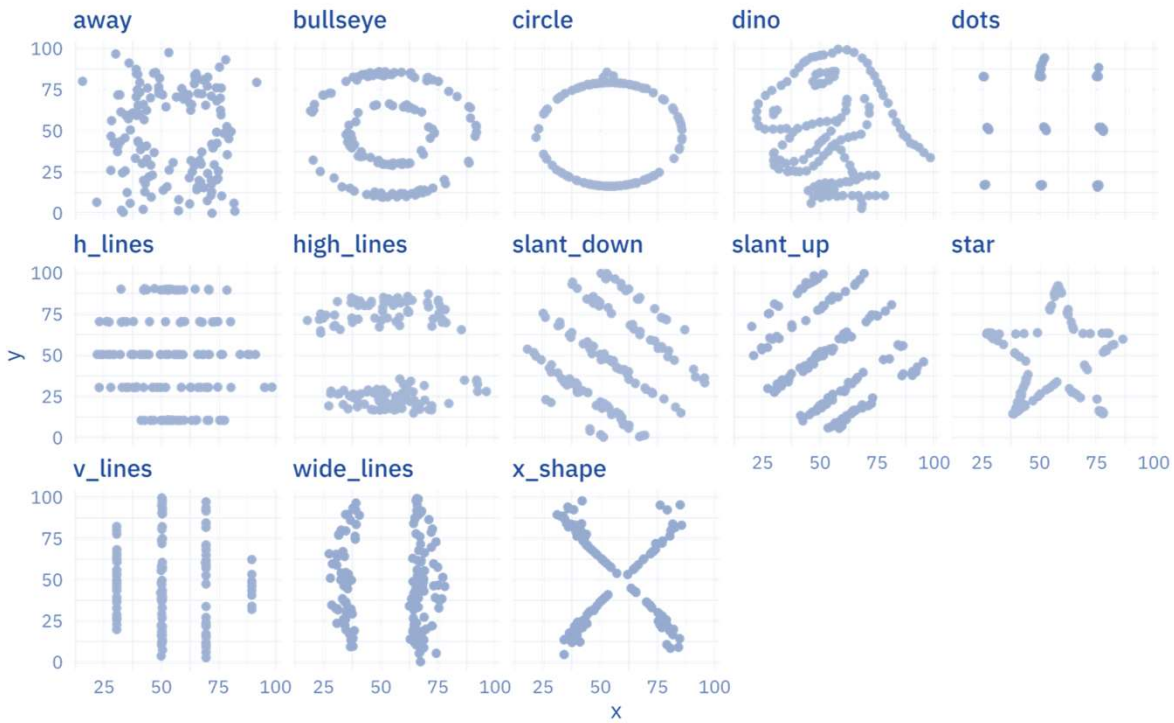
Another example



Common statistical values for each group in the dataset

	Summary statistics					*Regression results*	
Dataset	Mean x	Mean y	Std Dev x	Std Dev y	Corr x y	Intercept	Coefficients
Away	54.27	47.83	16.77	26.94	-0.06	53.43	-0.10
Bullseye	54.27	47.83	16.77	26.94	-0.07	53.81	-0.11
Circle	54.27	47.84	16.76	26.93	-0.07	53.80	-0.11
Dino	54.26	47.83	16.77	26.94	-0.06	53.45	-0.10
Dots	54.26	47.84	16.77	26.93	-0.06	53.10	-0.10
H_lines	54.26	47.83	16.77	26.94	-0.06	53.21	-0.10
High_lines	54.27	47.84	16.77	26.94	-0.07	53.81	-0.11
Slant_down	54.27	47.84	16.77	26.94	-0.07	53.85	-0.11
Slant_up	54.27	47.83	16.77	26.94	-0.07	53.81	-0.11
Star	54.27	47.84	16.77	26.93	-0.06	53.33	-0.10
V_lines	54.27	47.84	16.77	26.94	-0.07	53.89	-0.11
Wide_lines	54.27	47.83	16.77	26.94	-0.07	53.63	-0.11
X_shape	54.26	47.84	16.77	26.93	-0.07	53.55	-0.11

Descriptive statistics can be misleading. Data visualization helps.

Source: Datasaurus Dozen from wikipedia

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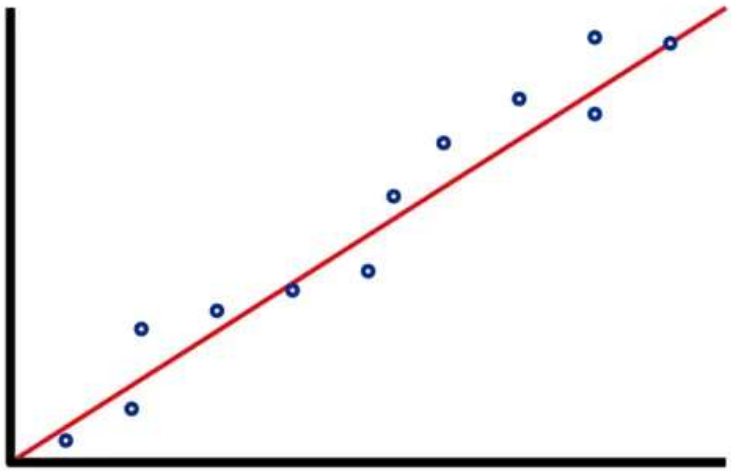
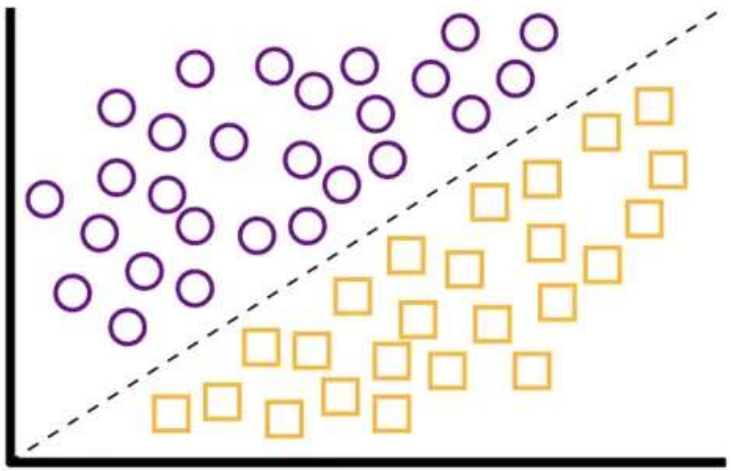
Classification vs Regression

Regression



Classification



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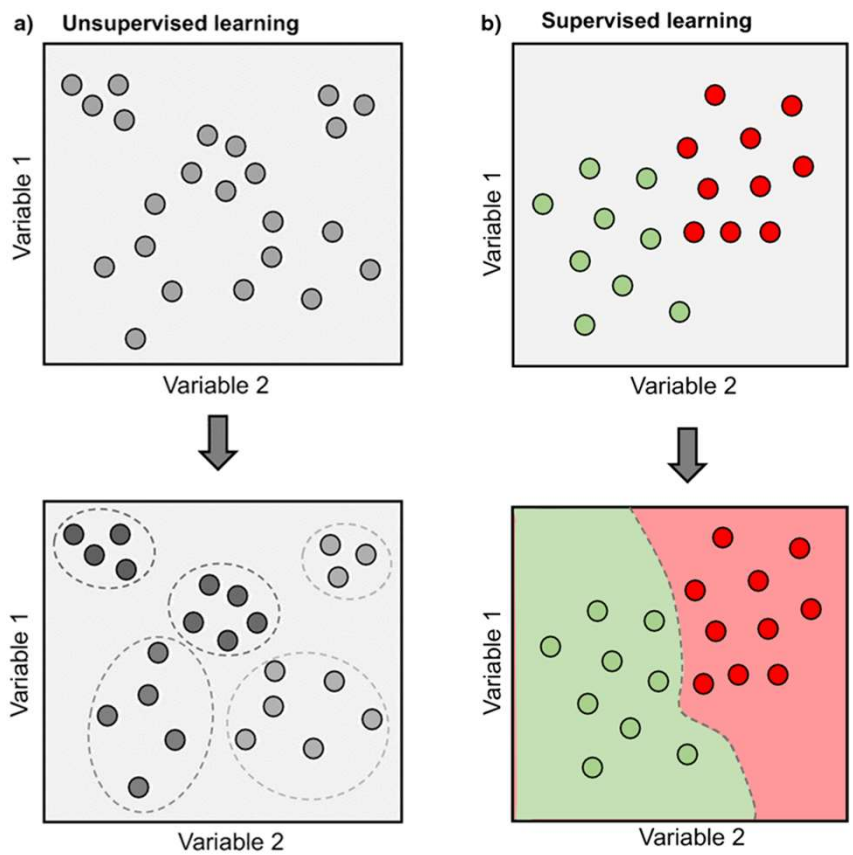
Unsupervised vs Supervised

Unsupervised learning
Input data is unlabeled
Has no feedback mechanism
Assigns properties of given data to classify it
Divided into Clustering & Association
Used for analysis
Algorithms include: k-means clustering, hierarchical clustering, apriori algorithm
A unknown number of classes



Supervised learning
Input data is labeled
Has a feedback mechanism
Data is classified based on the training dataset
Divided into Regression & Classification
Used for prediction
Algorithms include: decision trees, logistic regressions, support vector machine
A known number of classes

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Machine Learning Algorithms

Source: "Neural Network for Porosity Prediction in Carbonate Formation", Huda F. Al-Saad, 2023

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Artificial General Intelligence (AGI)

What is Intelligence?

There are many definitions, here is one:

"Intelligence measures an agent's ability to achieve goals in a wide range of environments." S. Legg and M. Hutter

Narrow AI vs AGI

NARROW AI	ARTIFICIAL GENERAL INTELLIGENCE (AGI)
■ AI focused on a specific, singular or limited task ■ Examples include image recognition, hyper-personalization, chatbots, predictive text ■ Trained on specific tasks by data scientists ■ Correlates questions or assignments to a specific data set to accomplish a task ■ No self-awareness, consciousness, ability to think	■ Not fully realized, with some developers questioning if it will be possible ■ Seeks machines that can handle a range of cognitive tasks with little oversight ■ The ability to learn, generalize, apply knowledge and plan for the future ■ Must consistently pass the Turing Test ■ Single, general intelligence that possesses common sense and creativity and expresses emotions

Source: Definition from "A Collection of Definitions of Intelligence", Shane Legg Marcus Hutter, 2007, Image from Tech target under fair use